

Results — revised submission — original style

1 Datasets

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	VirusPseAAC	207	440	6	8.62	0.92
3	CHD_49	555	49	6	5.77	1.60
4	Emotions	593	72	6	1.48	0.16
5	Scene	2407	294	6	1.25	0.11
6	PlantPseAAC	978	440	12	6.69	0.68
7	Water-quality	1060	16	14	1.77	0.29
8	Yeast	2417	103	14	7.20	1.82
9	HumanPseAAC	3106	440	14	15.29	1.05
10	enron	1702	1001	53	10.45	0.80

Table 1: (Table 3) Datasets used in the revised experiments. N , P , K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

Pending follow-ups (to fill before final submission)

- **LGBM run on enron.** Sequentially training (2–3h per task due to $K=53$ pairwise classifiers); will be merged into `full_enron_split_lgbm_summary/` when complete and the LGBM aggregate panels regenerated.
- **Extended PartialAbstention metrics.** The evaluator has been extended with 11 new PA metrics (`jaccard_pa`, `{example,macro,micro}-{precision,recall,f1}_pa`, `mfrd_pa`, `afrd_pa`). Existing runs were summarised with the previous metric set; the new columns will populate automatically on the next end-to-end training+evaluation pass.

2 Method encoding

For every panel that follows, the x-axis encodes the classifier as one of 12 methods, grouped into three blocks:

Block A — Order-based predictors (ours, indices 1–8). Each name has the form `<order>-<target>-<height>`; $order \in \{PA = \text{partial order}, PR = \text{pre-order}\}$; $target \in \{H = \text{Hamming}, S = \text{Subset } 0/1\}$; $height \in \{2, \infty \text{ (omitted in the label)}\}$.

- **1 = PA-H-2:** partial order, Hamming target, height ≤ 2 .

- 2 = **PA-H**: partial order, Hamming target, unrestricted height.
- 3 = **PA-S-2**: partial order, Subset 0/1 target, height ≤ 2 .
- 4 = **PA-S**: partial order, Subset 0/1, unrestricted.
- 5 = **PR-H-2**: pre-order, Hamming, height ≤ 2 .
- 6 = **PR-H**: pre-order, Hamming, unrestricted.
- 7 = **PR-S-2**: pre-order, Subset 0/1, height ≤ 2 .
- 8 = **PR-S**: pre-order, Subset 0/1, unrestricted.

Block B — Standard MLC baselines from the original paper (indices 9–11).

- 9 = **BR** (Binary Relevance): K independent binary classifiers, one per label.
- 10 = **CC** (Classifier Chain): sequential chain that uses previous labels as features.
- 11 = **CLR** (Calibrated Label Ranking): pairwise label ranking with a calibration label that separates relevant from irrelevant.

Block C — Extended baseline added for the revision (index 12).

- 12 = **ECC** (Ensemble of Classifier Chains): averages over multiple CCs with random chain orders.

Reading the panels. Each panel plots one dotted line with markers per noise level, colored as in the original paper’s Table 5: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$, $\alpha = 0.3$. *PartialAbstention* and *ScoreVector* panels show only indices 1–8 (the remaining baselines do not export a partial-abstention prediction nor a probabilistic score vector). Vertical dashed lines at $x = 4.5$, $x = 8.5$, and $x = 11.5$ separate the four method groups: PA (1–4), PR (5–8), standard baselines BR/CC/CLR (9–11), and ECC (12). Missing methods show as gaps in the line (with a small grey “x” on the axis when the method has no data at all for that panel).

3 Paper results (RF base learner)

Reproduces the layout of the original paper’s main experiment tables (Tables 5–7): 3 datasets across, 4 metrics down, one table per prediction type (binary vector, partial abstention, score vector). Per the revision, $f_{\text{MLC}}^{\text{ham}}$ and $f_{\text{MLC}}^{\text{sub}}$ are dropped in favour of $f_{\text{MLC}}^{\text{Jacc}}$.

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
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10	enron	1702	1001	53	10.45	0.80

Table 2: Datasets (repeated from Table 1 for reference). N , P , K : instances, features, labels. MeanIR and CVIR: per-label imbalance (higher = more imbalanced).

3.1 Standard MLC predictions (BinaryVector)

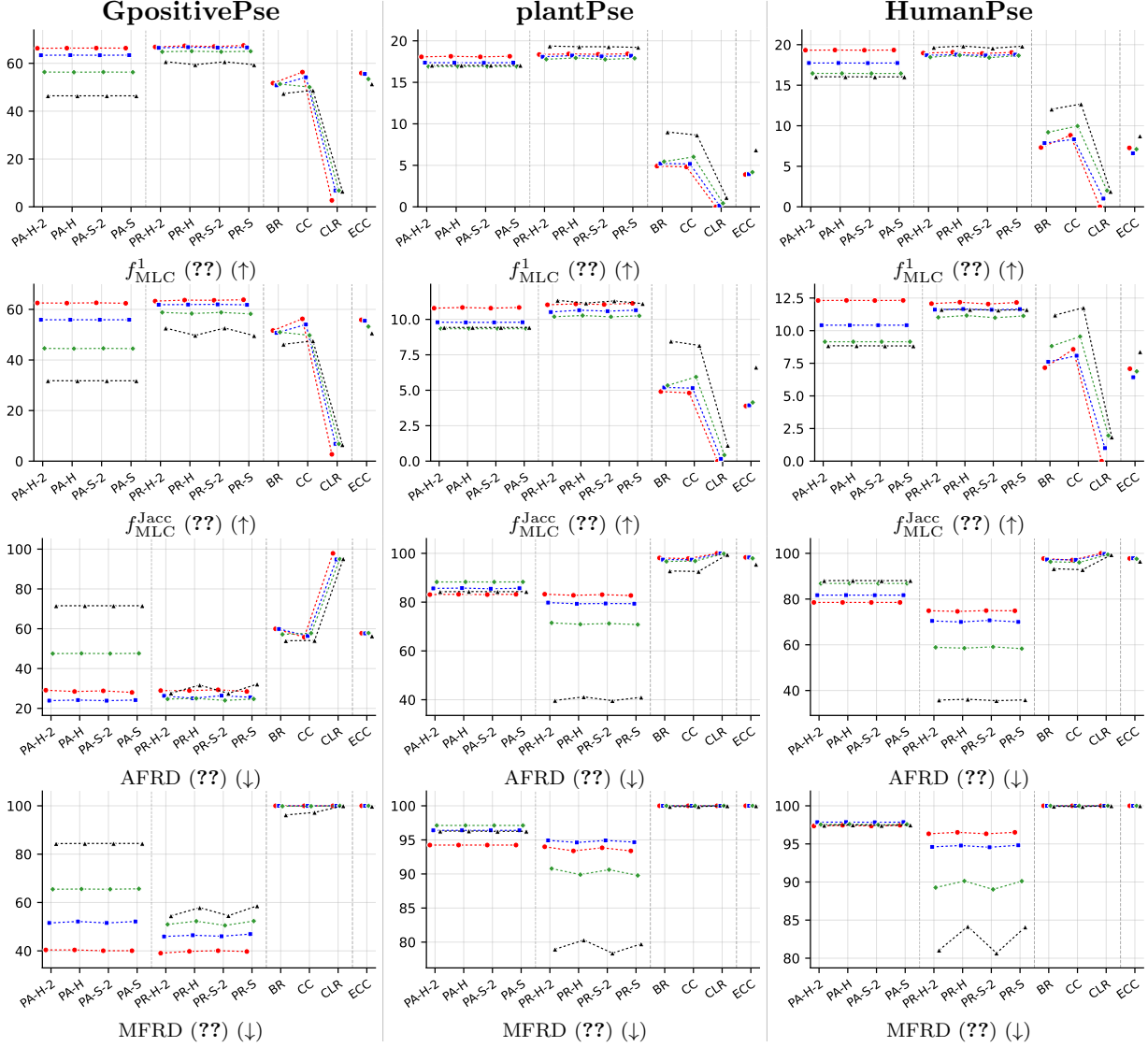


Table 3: (Table 4) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the classifiers (x -axis): partial-order predictors (PA-H-2, PA-H, PA-S-2, PA-S), preorder predictors (PR-H-2, PR-H, PR-S-2, PR-S), and standard MLC baselines (BR, CC, CLR). Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.2 Partial abstention (PartialAbstention)

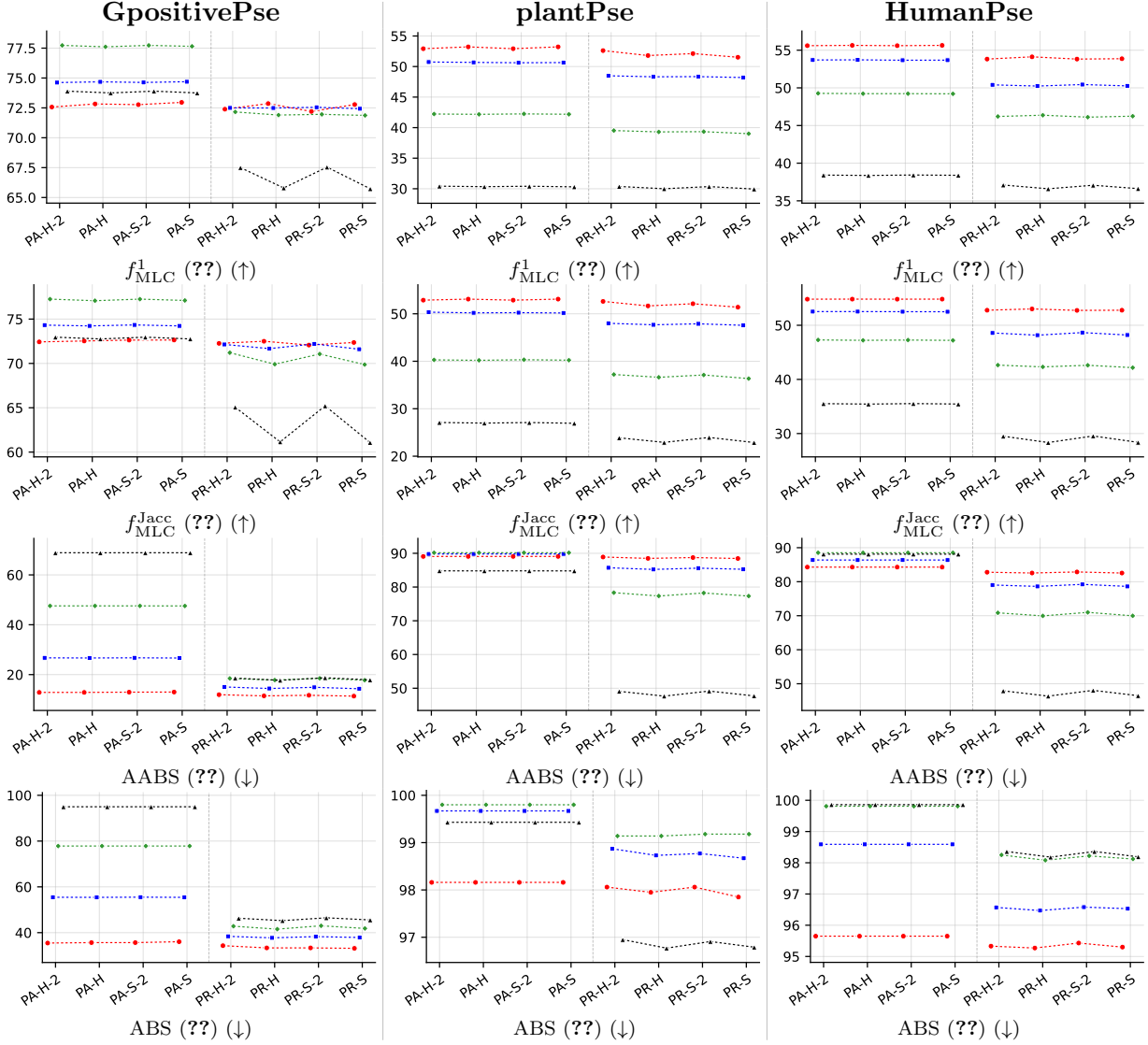
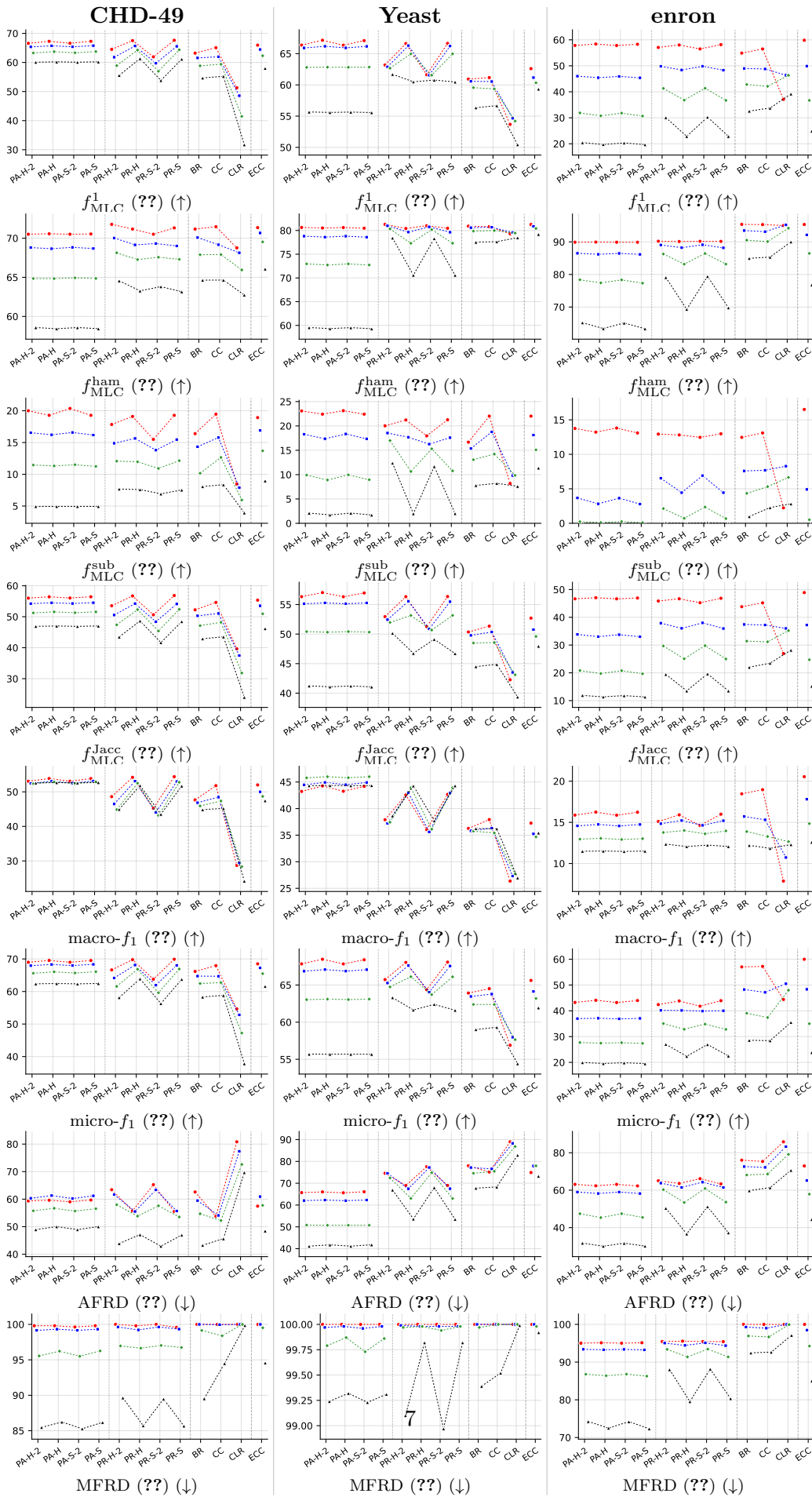


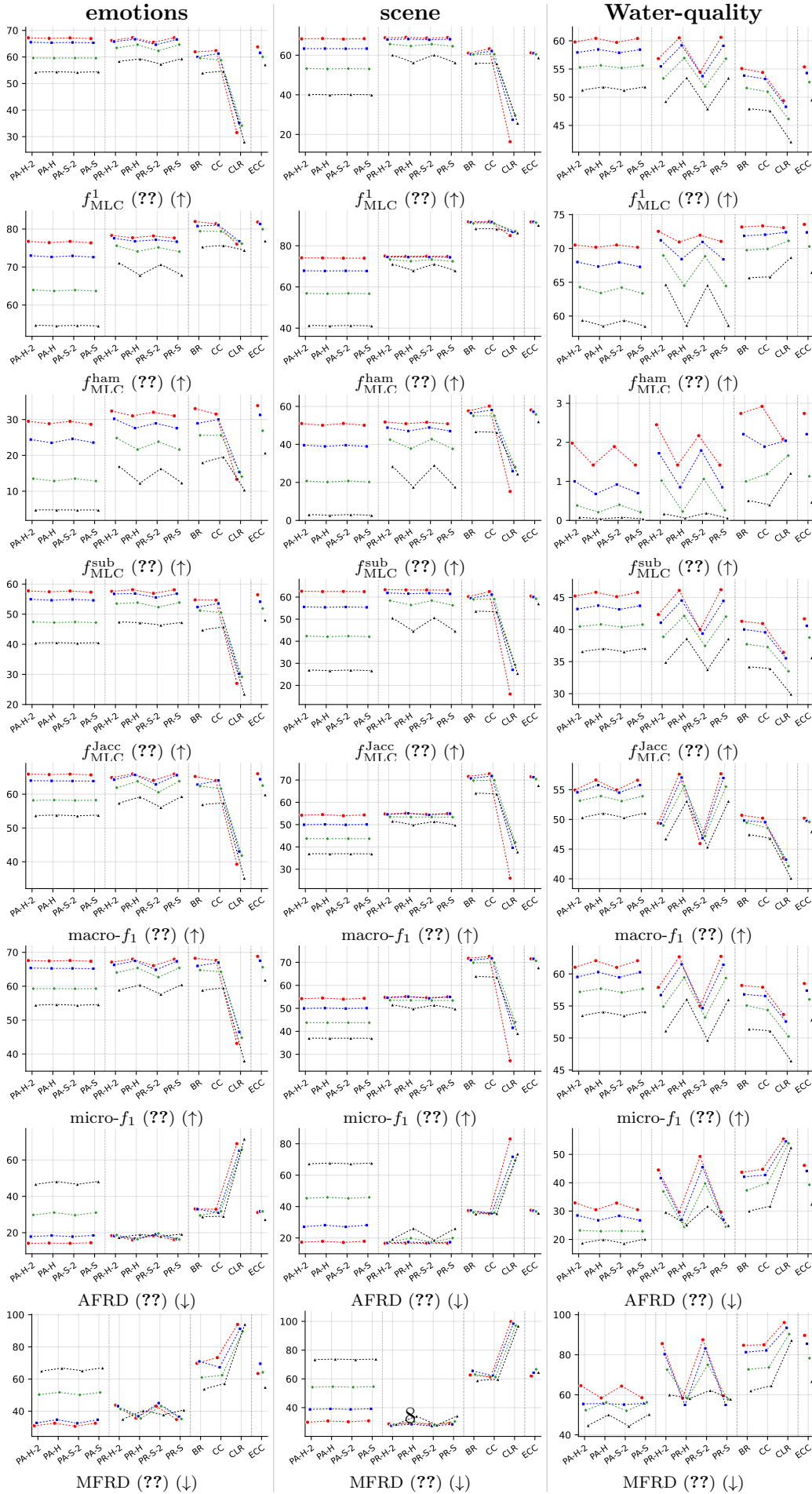
Table 4: (Table 5) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the classifiers (x -axis): partial-order predictors (PA-H-2, PA-H, PA-S-2, PA-S) and preorder predictors (PR-H-2, PR-H, PR-S-2, PR-S). Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.3 Experimental results — appendix tables (RF base learner)

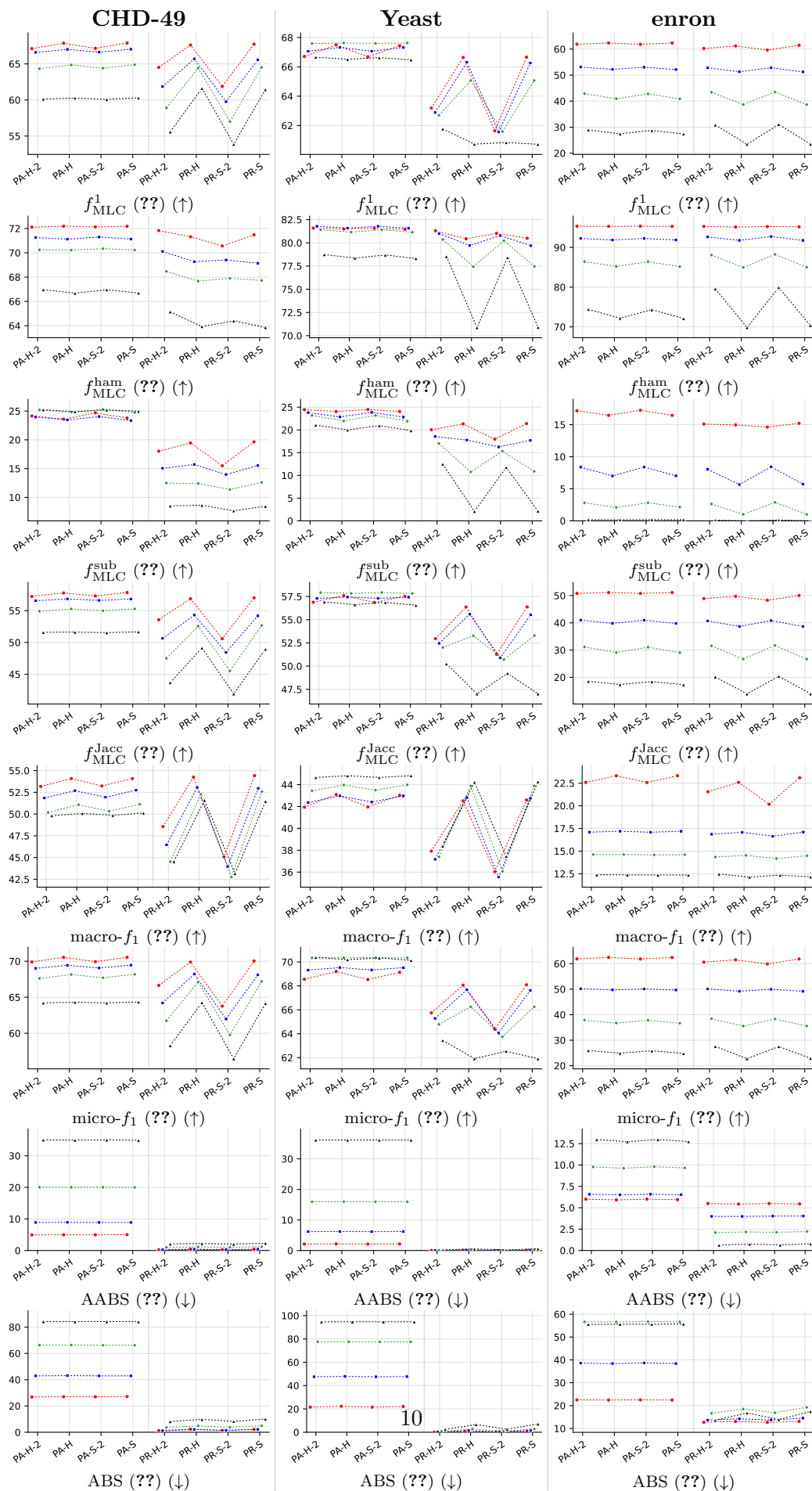
Appendix counterpart of the main Paper results section. Tables All tables share the same 8-metric set (original $5 + 3$ revision additions: f_{MLC}^{Jacc} , macro- f_1 , micro- f_1). G2/G4 = imbalanced datasets, G3/G5 = balanced datasets (split follows the original paper’s appendix). G6/G7 = main-paper datasets (GpositivePse, PlantPse, HumanPse) shown with the full metric set; the main body only carries the revised compact Tables 4/5.

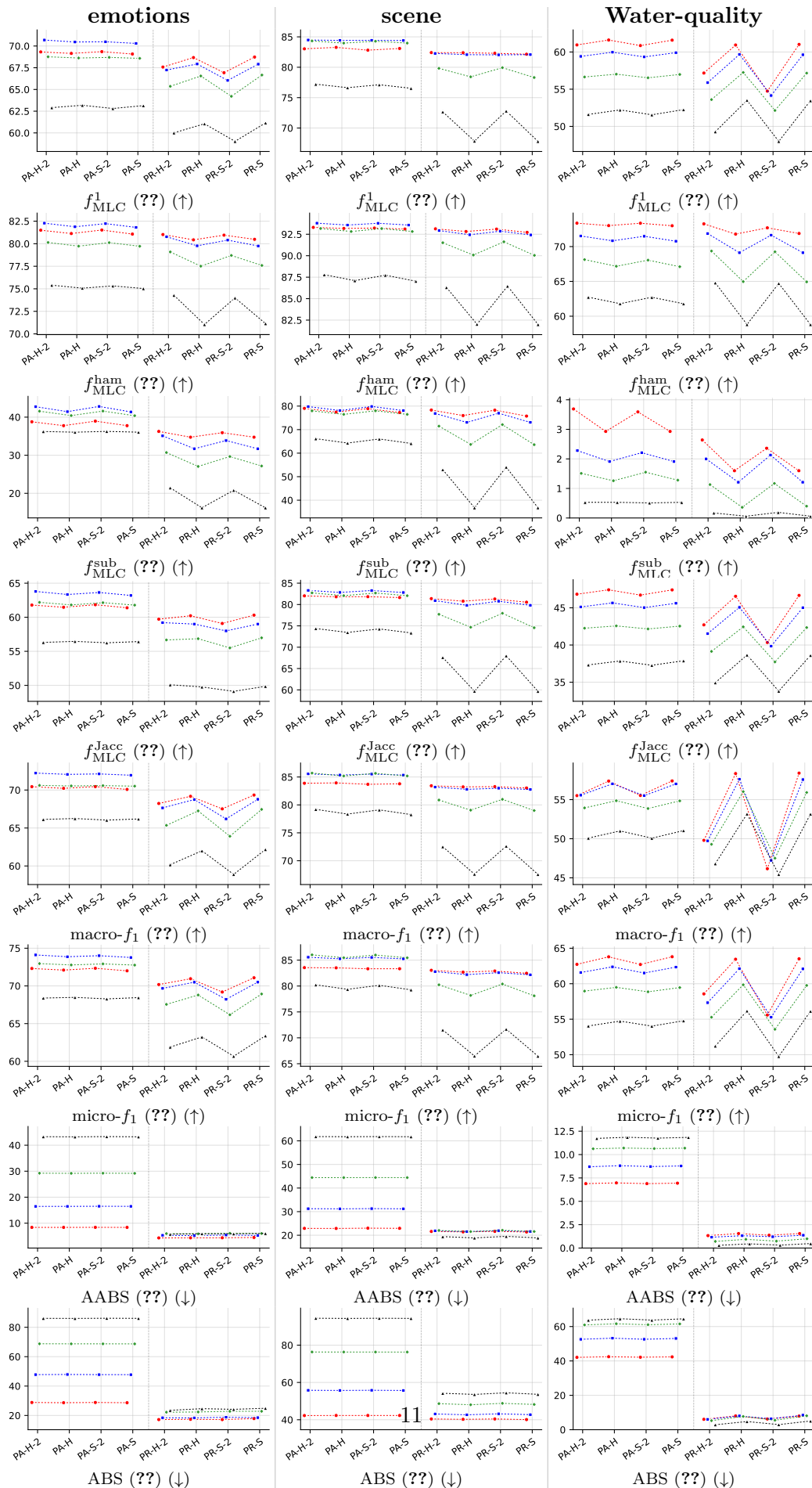
3.3.1 Standard MLC predictions (BinaryVector)





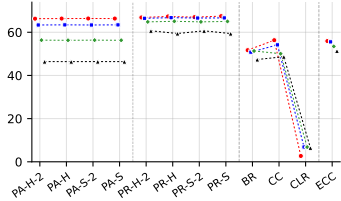
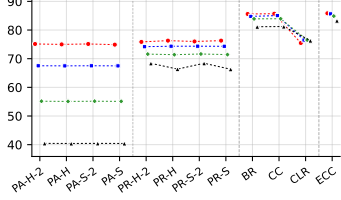
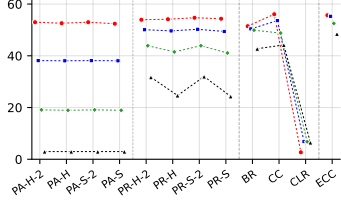
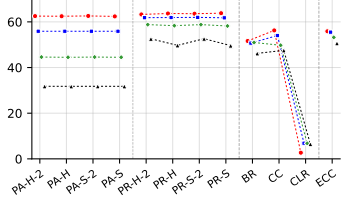
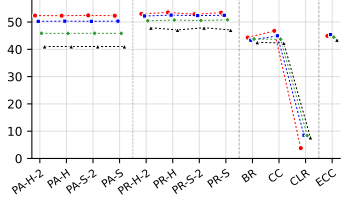
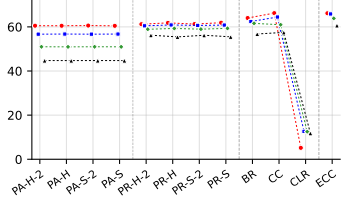
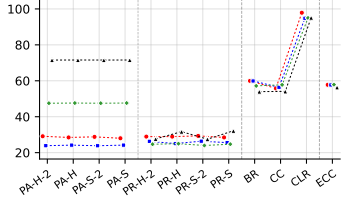
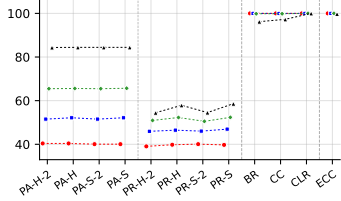
3.3.2 Partial abstention (PartialAbstention)



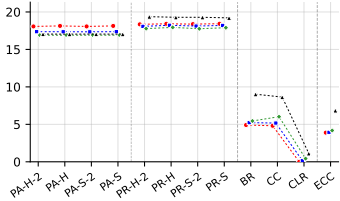
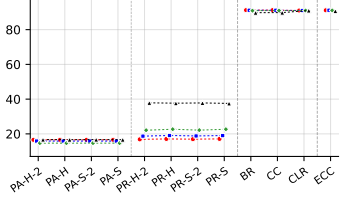
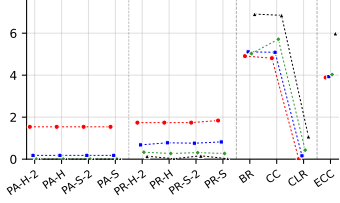
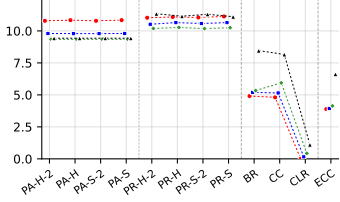
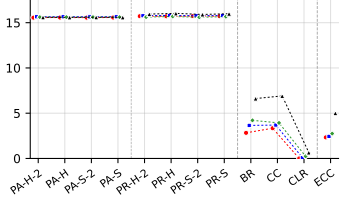
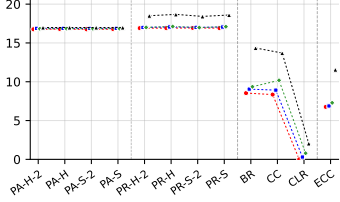
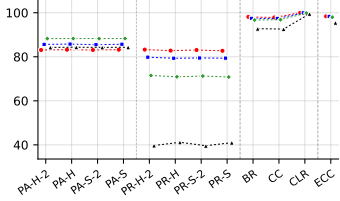
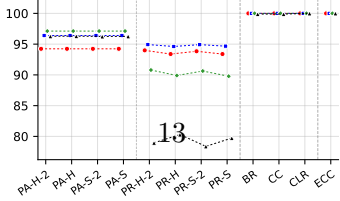


3.3.3 Standard MLC predictions (BinaryVector)

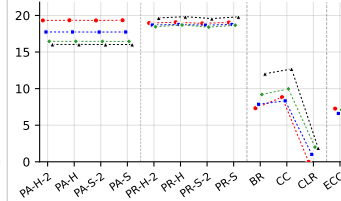
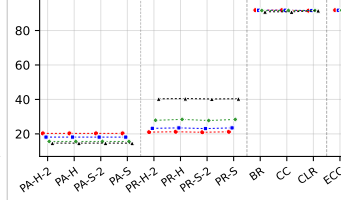
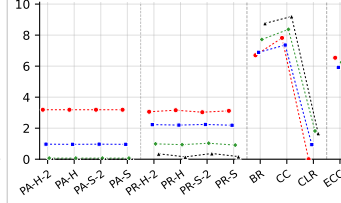
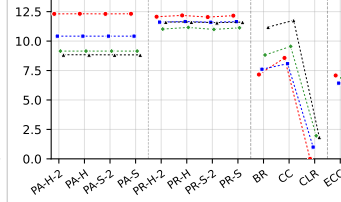
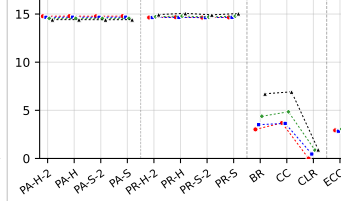
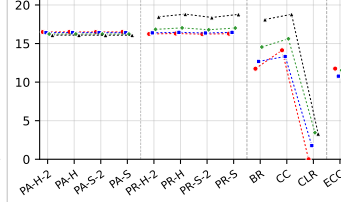
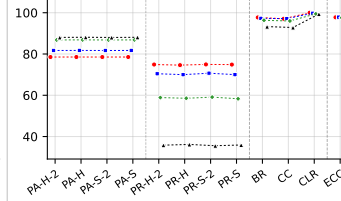
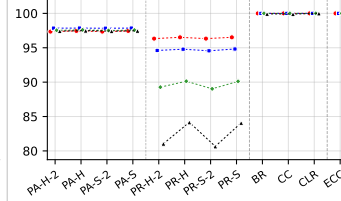
GpositivePse

 f_{MLC}^1 (??) ($\uparrow$) f_{MLC}^{ham} (??) ($\uparrow$) f_{MLC}^{sub} (??) ($\uparrow$) f_{MLC}^{Jacc} (??) ($\uparrow$)macro- f_1 (??) ($\uparrow$)micro- f_1 (??) ($\uparrow$)AFRD (??) ($\downarrow$)MFRD (??) ($\downarrow$)

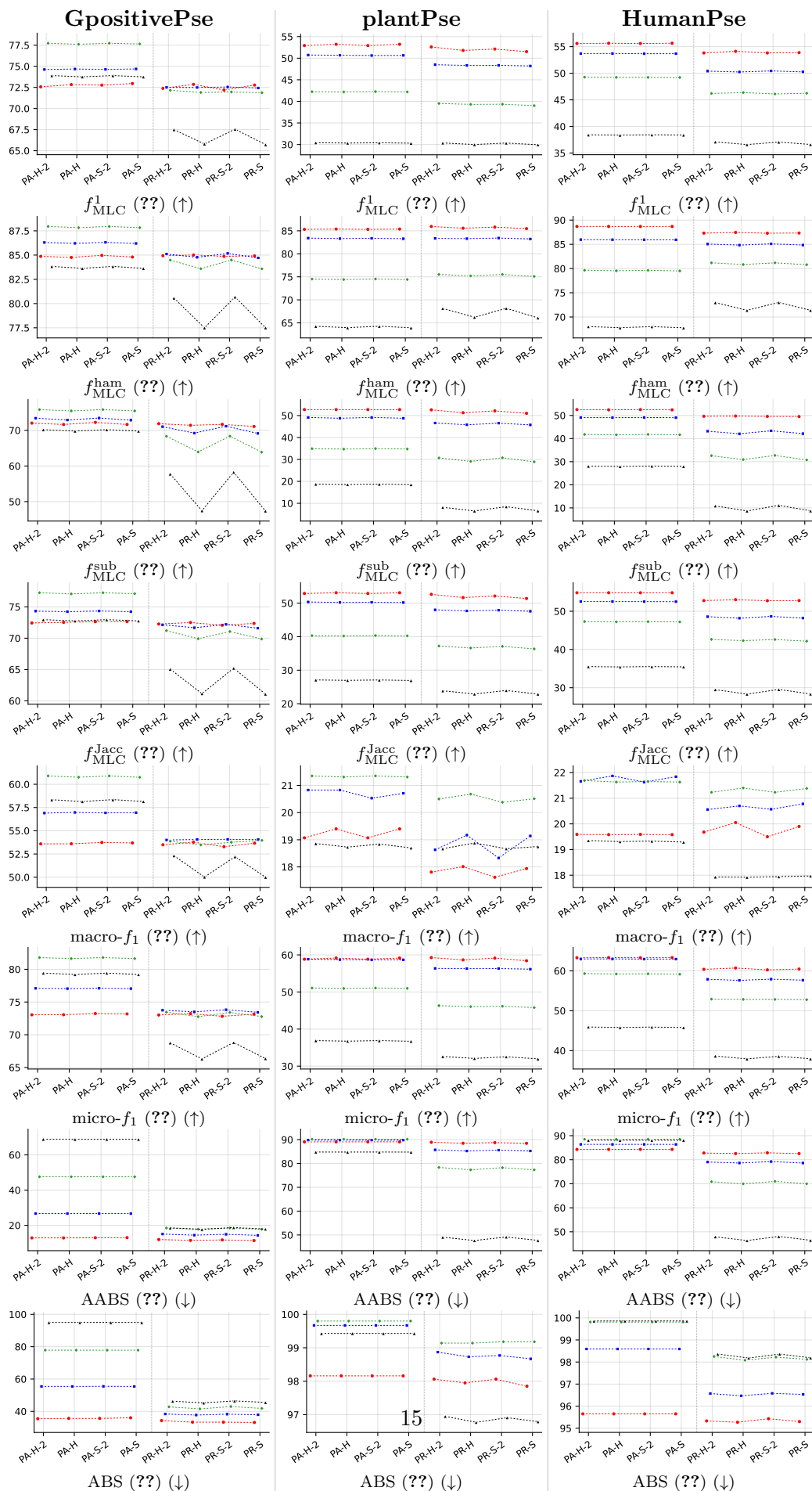
plantPse

 f_{MLC}^1 (??) ($\uparrow$) f_{MLC}^{ham} (??) ($\uparrow$) f_{MLC}^{sub} (??) ($\uparrow$) f_{MLC}^{Jacc} (??) ($\uparrow$)macro- f_1 (??) ($\uparrow$)micro- f_1 (??) ($\uparrow$)AFRD (??) ($\downarrow$)MFRD (??) ($\downarrow$)

HumanPse

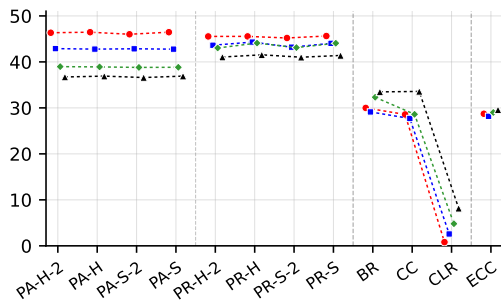
 f_{MLC}^1 (??) ($\uparrow$) f_{MLC}^{ham} (??) ($\uparrow$) f_{MLC}^{sub} (??) ($\uparrow$) f_{MLC}^{Jacc} (??) ($\uparrow$)macro- f_1 (??) ($\uparrow$)micro- f_1 (??) ($\uparrow$)AFRD (??) ($\downarrow$)MFRD (??) ($\downarrow$)

3.3.4 Partial abstention (PartialAbstention)

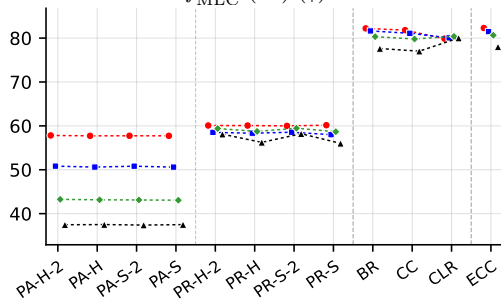


3.3.5 Standard MLC predictions (BinaryVector)

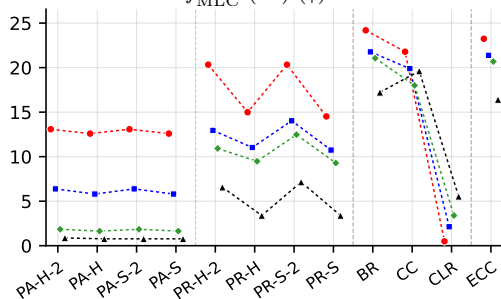
VirusPseAAC



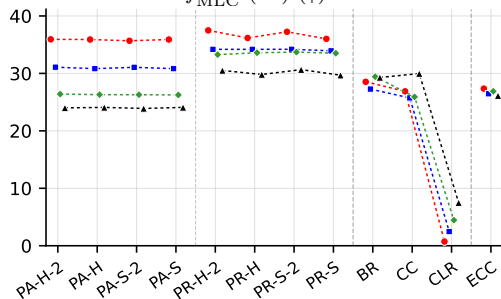
f_{MLC}^1 (??) (↑)



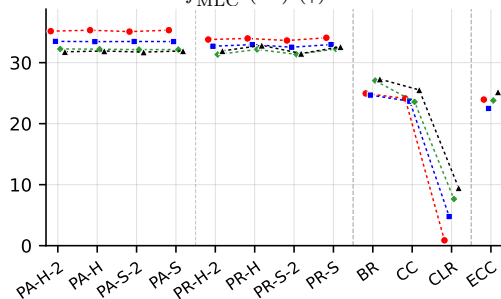
f_{MLC}^{ham} (??) (↑)



f_{MLC}^{sub} (??) (↑)



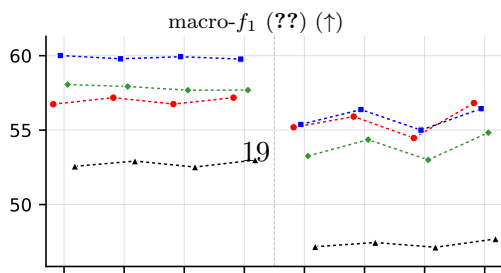
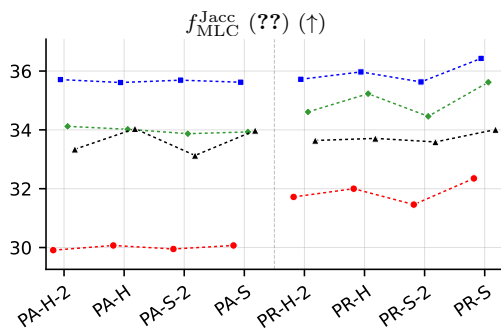
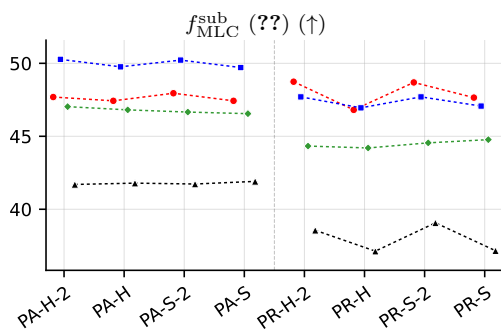
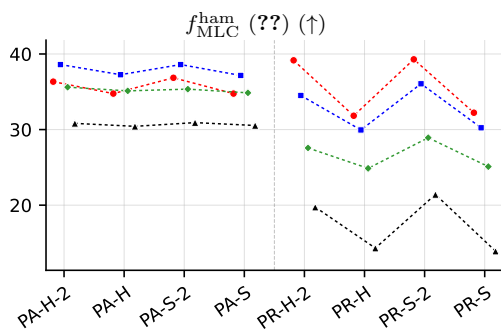
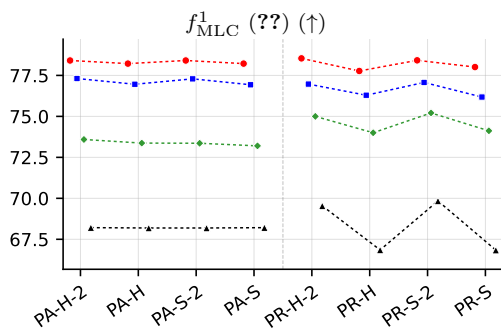
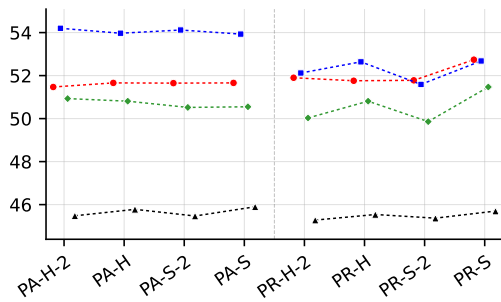
f_{MLC}^{Jacc} (??) (↑)



$macro-f_1$ (??) (↑)

3.3.6 Partial abstention (PartialAbstention)

VirusPseAAC



3.4 Abstention overview: aggregate trends

Purpose. This compact section shows the headline abstention-vs-standard-MLC trend across noise levels, averaged over all datasets, for both base learners (RF, LGBM) and both retained quality metrics (f_1 , Jaccard). Use it as a quick visual summary before diving into the detailed per-method / per-dataset figures in Section ??.

How to read. In each panel: blue bar = standard MLC (no abstention) on its native quality metric; coloured bar = with-abstention quality on retained labels. Green “ $+x.y$ ” label = score-point gain of abstention over the standard-MLC bar of the same method. The four sub-panels per figure correspond to $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

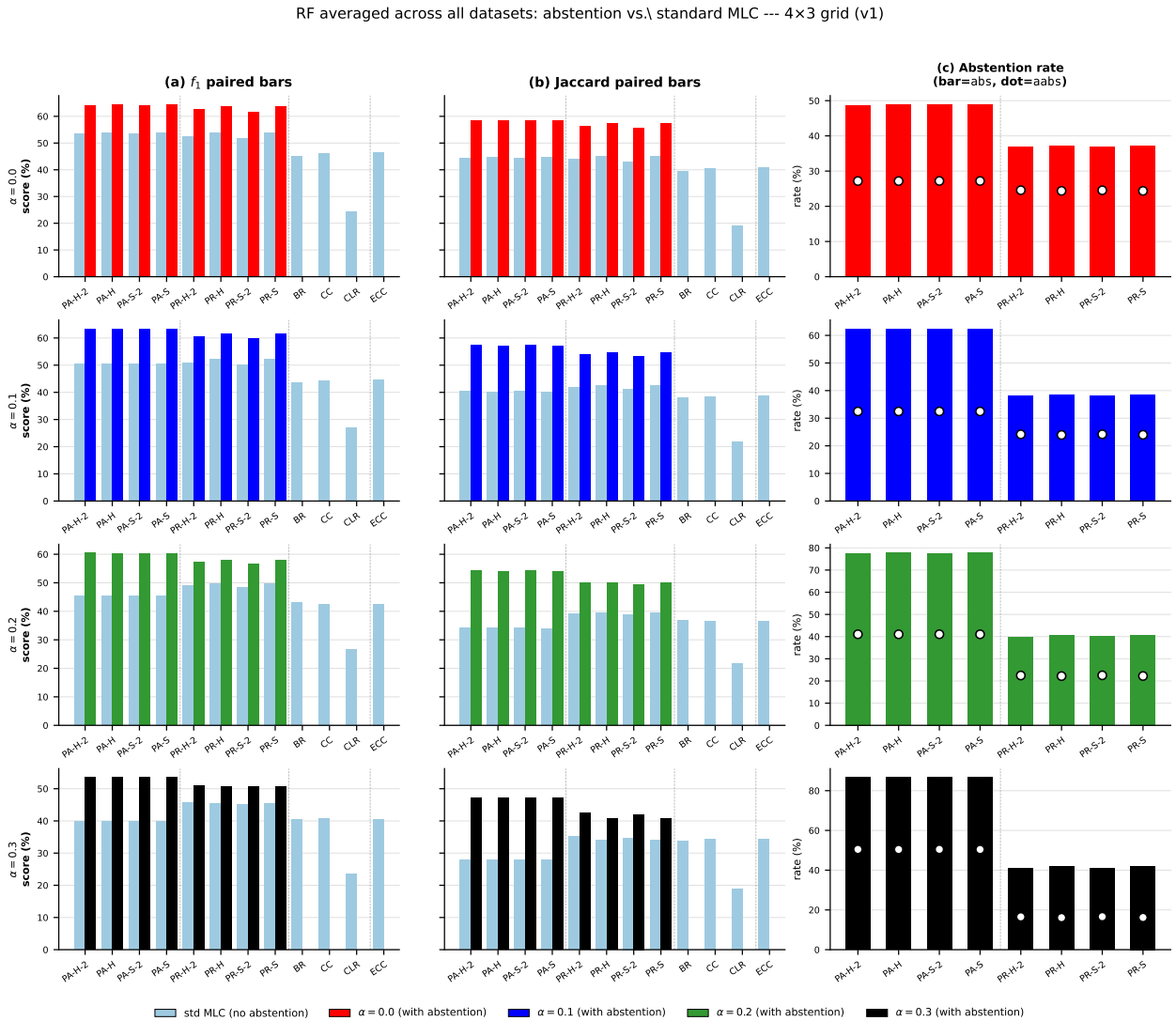


Figure 1: RF averaged across all datasets, 4×3 grid (rows = α , cols = f_1 / Jaccard / abstention rate) — (version 1) all 12 methods on the x -axis (PA + PR + baselines BR/CC/CLR/ECC); baselines plot only the std-MLC bar since they cannot abstain. Light blue = standard MLC, α -coloured = with-abstention; in (c), bar = **abs** (instance-level), dot = **aabs** (per-cell). Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

RF averaged across all datasets: abstention vs. standard MLC --- 4×3 grid (v2, 8 methods + gains)

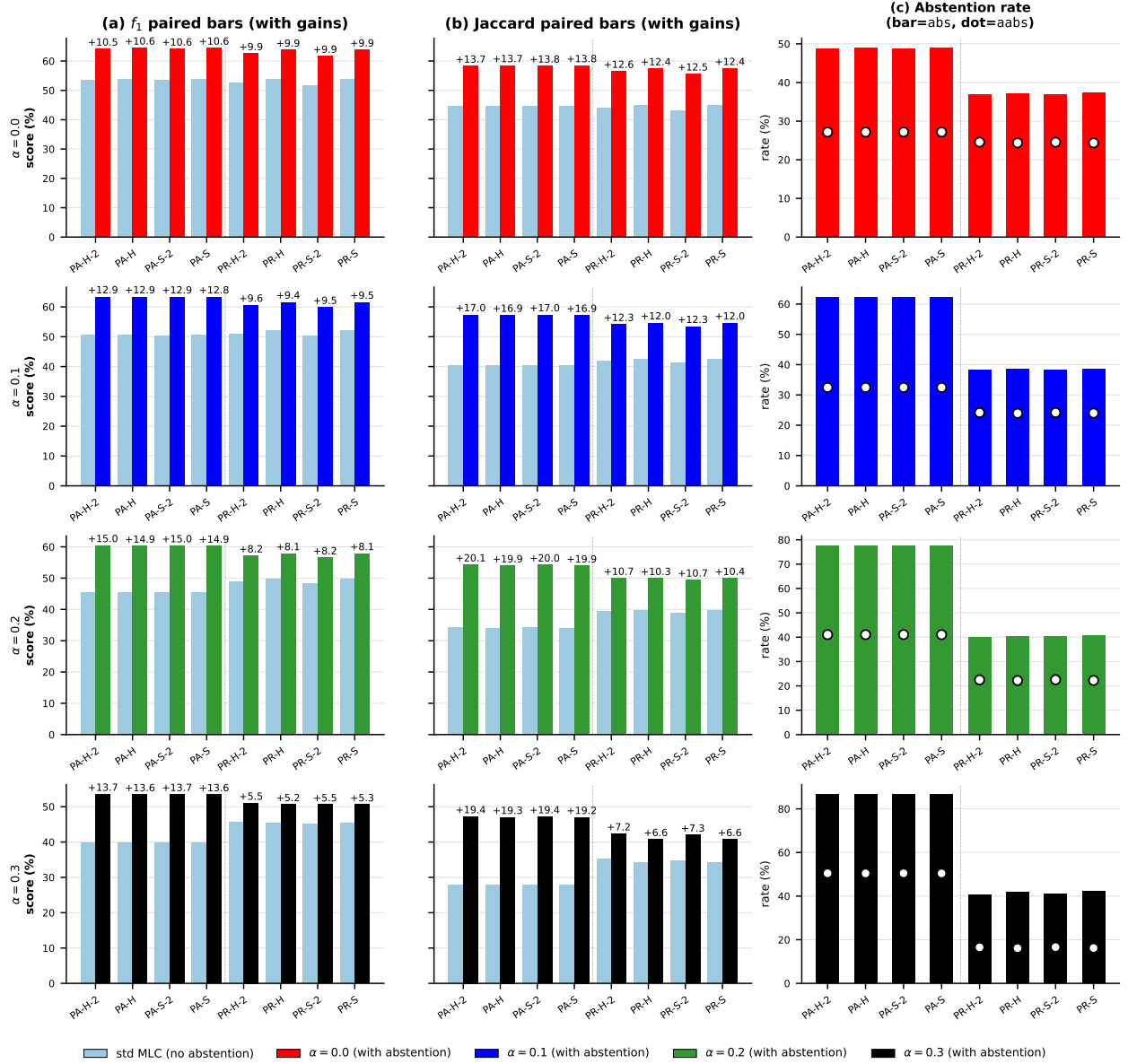


Figure 2: RF averaged across all datasets, 4 × 3 grid (rows = α , cols = f_1 / Jaccard / abstention rate) — (version 2) only the 8 PA/PR methods that can abstain; gain labels + $x.y$ printed above each with-abstention bar so the score-point benefit over the std-MLC bar at the same method is visible at a glance. Same colour / layout conventions as version 1. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

LGBM averaged across all datasets: abstention vs. standard MLC --- 4×3 grid (v1)

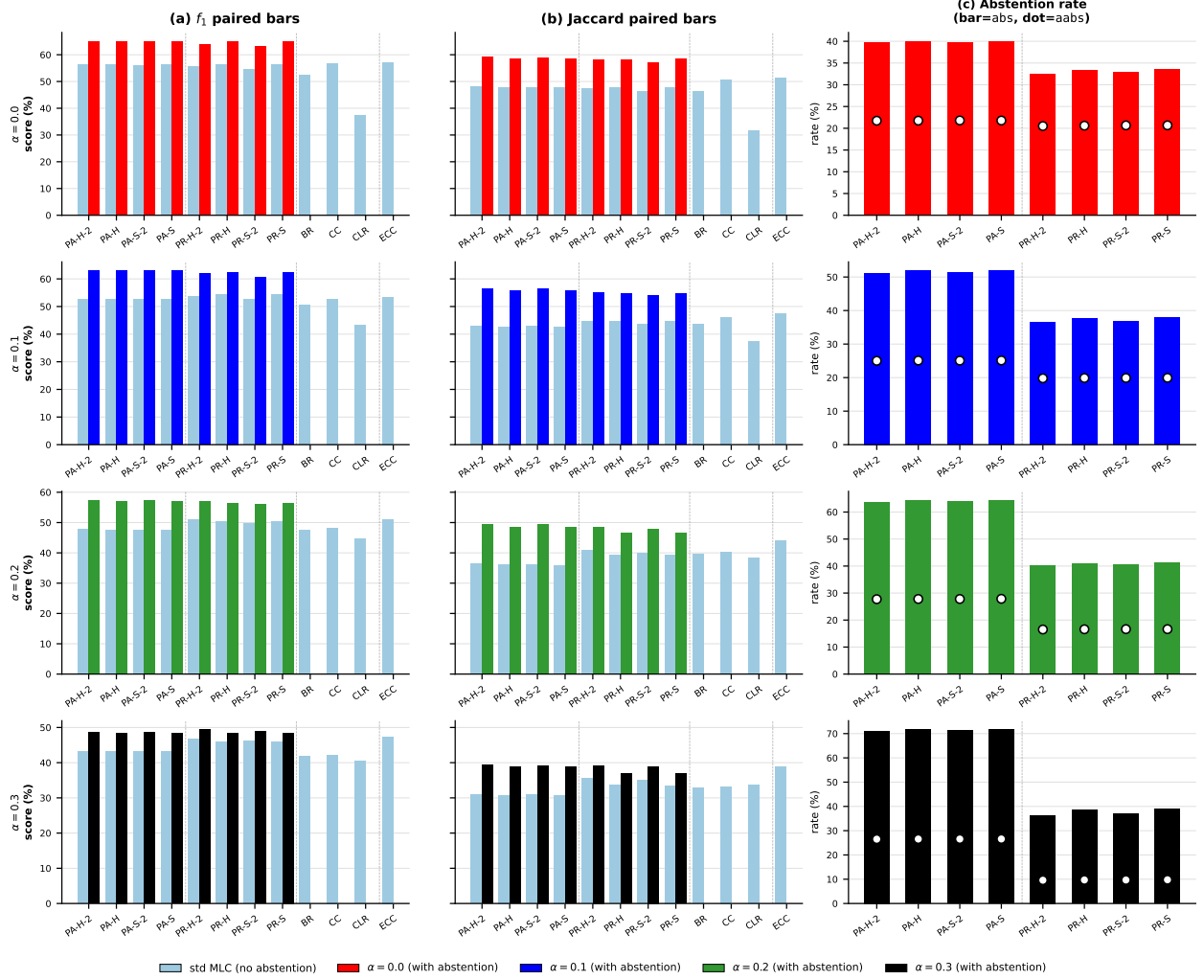


Figure 3: LGBM averaged across all datasets, 4×3 grid (rows = α , cols = f_1 / Jaccard / abstention rate) — (version 1) all 12 methods on the x -axis (PA + PR + baselines BR/CC/CLR/ECC); baselines plot only the std-MLC bar since they cannot abstain. Light blue = standard MLC, α -coloured = with-abstention; in (c), bar = abs (instance-level), dot = aabs (per-cell). Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

LGBM averaged across all datasets: abstention vs. standard MLC --- 4×3 grid (v2, 8 methods + gains)

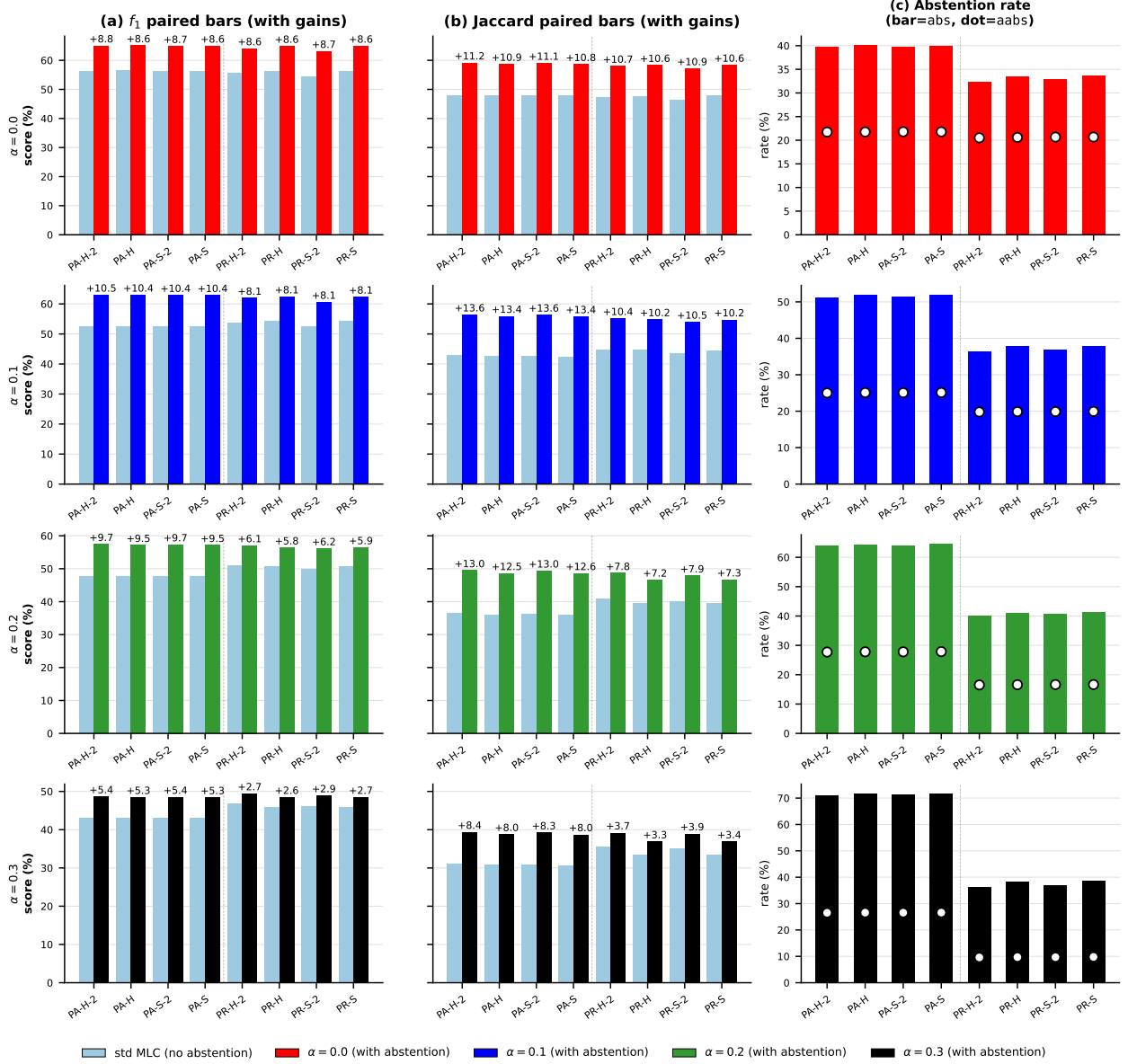


Figure 4: LGBM averaged across all datasets, 4×3 grid (rows = α , cols = f_1 / Jaccard / abstention rate) — (version 2) only the 8 PA/PR methods that can abstain; gain labels + $x.y$ printed above each with-abstention bar so the score-point benefit over the std-MLC bar at the same method is visible at a glance. Same colour / layout conventions as version 1. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.